Asymptotic Analysis: Big-O and Little-o

Landau notation, asymptotic equivalence, and their algebra, for calculus and real analysis

This document was generated by a large language model (LLM) and may contain errors.

1 Setup

Throughout, f and g are real-valued functions defined near a fixed base point a, where a may be a finite real number or $\pm\infty$. "Near a" means on some deleted neighborhood of a if $a \in \mathbb{R}$, or on some ray (M, ∞) or $(-\infty, M)$ if $a = \pm\infty$. All statements below are local: a function can be $O(g)$ as $x \to 0$ and something completely different as $x \to \infty$, so the base point is always part of the claim, even when left implicit.

Note. When a limit definition below divides by $g(x)$, assume $g(x) \neq 0$ near a (except possibly at a itself).

2 The Landau symbols

Definition 2.1 (Little-o)**.** $f(x) = o\big(g(x)\big)$ as $x \to a$ means: for every $\varepsilon > 0$ there is a neighborhood of a on which

$$|f(x)| \leq \varepsilon\,|g(x)|.$$

Equivalently, if g is eventually nonzero near a,

$$\lim_{x \to a} \frac{f(x)}{g(x)} = 0.$$

Read "f is of strictly smaller order than g near a," or "f is negligible compared to g."

Definition 2.2 (Big-O)**.** $f(x) = O\big(g(x)\big)$ as $x \to a$ means: there exist $C > 0$ and a neighborhood of a on which

$$|f(x)| \leq C\,|g(x)|.$$

Equivalently, if g is eventually nonzero near a, the ratio $f(x)/g(x)$ is *bounded* near a — it need not converge. Read "f is of order at most g near a."

Definition 2.3 (Big-Θ)**.** $f(x) = \Theta\big(g(x)\big)$ as $x \to a$ means $f = O(g)$ and $g = O(f)$: there exist $c, C > 0$ with

$$c\,|g(x)| \leq |f(x)| \leq C\,|g(x)|$$

near a. Read "f and g have the same order" — a symmetric relation, unlike O and o.

Definition 2.4 (Asymptotic equivalence)**.** $f(x) \sim g(x)$ as $x \to a$ means

$$\lim_{x \to a} \frac{f(x)}{g(x)} = 1.$$

This is strictly stronger than Θ: it fixes the constant of proportionality to be exactly 1 in the limit, not merely bounded above and below.

Remark 2.5. These are statements about *one function belonging to a set of functions*: $O(g)$ denotes the set of all f satisfying the bound above, and "$f = O(g)$" is shorthand for "$f \in O(g)$." The equals sign is an abuse of notation inherited from long-standing convention, not a genuine equality — it is not symmetric ($x = O(x^2)$ as $x \to 0$, but writing $O(x^2) = x$ is meaningless), and two different functions can both equal $O(g)$ without equaling each other.

3 Basic algebra

Fix the base point a. These hold for functions compared near a.

- **o implies O.** If $f = o(g)$ then $f = O(g)$ (take $C = 1$ and shrink the neighborhood). The converse fails: $f = O(g)$ does not imply $f = o(g)$.
- **Transitivity.** $f = O(g)$ and $g = O(h)$ imply $f = O(h)$; likewise for o, and for mixed chains ($f = o(g)$, $g = O(h) \Rightarrow f = o(h)$).
- **Constant multiples are absorbed.** $O(cg) = O(g)$ and $o(cg) = o(g)$ for any constant $c \neq 0$. In particular $c \cdot O(g) = O(g)$.
- **Sum rule (absorption).** $O(g) + O(g) = O(g)$, and more generally $O(g) + O(h) = O(\max(|g|, |h|))$. The same holds for o. A sum is governed by its dominant term: $o(g) + O(g) = O(g)$.
- **Product rule.** $O(f) \cdot O(g) = O(fg)$, and $o(f) \cdot O(g) = o(fg)$. Multiplying two little-o terms gives little-o of the product: $o(f) \cdot o(g) = o(fg)$.
- **Powers.** If $f = O(g)$ with $f, g > 0$ near a, then $f^k = O(g^k)$ for $k > 0$; similarly for o.
- **No cancellation.** $O(g) - O(g)$ is *not* 0; it is again $O(g)$. Landau symbols name an upper bound on size, not a specific function, so "subtracting" two unspecified bounded quantities does not cancel them. Never manipulate an O- or o-expression as if it were an ordinary algebraic term standing for one fixed function.
- **Composition is not automatic.** $f = O(g)$ as $x \to a$ does *not* generally give $h(f(x)) = O(h(g(x)))$ for an arbitrary h; this needs extra hypotheses on h (e.g. h Lipschitz, or monotone with controlled growth).

4 Taylor's theorem in Landau notation

Landau notation is the natural language for truncating a Taylor expansion: it records how fast the error shrinks without writing the error term explicitly.

Theorem 4.1 (Peano remainder)**.** *If f is n times differentiable at a, then as $x \to a$,*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big).$$

Theorem 4.2 (Lagrange / big-O remainder)**.** *If f is $(n+1)$ times differentiable on an interval containing a and x, with the $(n+1)$-st derivative bounded there, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + O\big((x-a)^{n+1}\big)$$

as $x \to a$ — one order sharper than the Peano form, at the cost of the stronger smoothness hypothesis.

Example 4.3 (Standard expansions as $x \to 0$)**.**

$$\begin{aligned}
e^x &= 1 + x + \frac{x^2}{2} + O(x^3) \\
\sin x &= x - \frac{x^3}{6} + o(x^3) \\
\cos x &= 1 - \frac{x^2}{2} + O(x^4) \\
\ln(1+x) &= x - \frac{x^2}{2} + O(x^3) \\
(1+x)^r &= 1 + rx + \binom{r}{2} x^2 + o(x^2) \qquad (r \in \mathbb{R})
\end{aligned}$$

Remark 4.4. The exponent inside the Landau term is the order of the *first omitted* term, not the number of terms kept. Writing $o(x^3)$ after a cubic term and $O(x^3)$ after a quadratic term are both legitimate but say different things about where the next correction enters.

5 Asymptotic equivalence: the finer tool

- $f \sim g$ as $x \to a$ *iff* $f - g = o(g)$, i.e. $f = g + o(g) = g\big(1 + o(1)\big)$.
- **Products and quotients behave well.** If $f_1 \sim g_1$ and $f_2 \sim g_2$, then $f_1 f_2 \sim g_1 g_2$ and $f_1/f_2 \sim g_1/g_2$. Powers follow too: $f^r \sim g^r$ for fixed r (where defined).
- **Sums do not.** $f_1 \sim g_1$ and $f_2 \sim g_2$ does *not* imply $f_1 + f_2 \sim g_1 + g_2$ — the leading terms can cancel. Example, as $x \to \infty$: take $g_1(x) = x$, $f_1(x) = x + 1$ (so $f_1 \sim g_1$), and $g_2(x) = -x$, $f_2(x) = -x + \sqrt{x}$ (so $f_2 \sim g_2$, since $f_2/g_2 = 1 - 1/\sqrt{x} \to 1$). Then $g_1 + g_2 \equiv 0$, while $f_1 + f_2 = 1 + \sqrt{x} \to \infty$ — so $f_1 + f_2$ cannot be asymptotically equivalent to $g_1 + g_2$. *Asymptotic equivalence is safe to use inside products and quotients, but never inside a sum or difference without checking that the leading terms survive cancellation.*
- **Equivalence controls the limit of ratios, not the size of the difference.** $f \sim g$ says nothing about $f - g \to 0$; it only says $f - g$ is small *relative to* g. E.g. $x^2 + x \sim x^2$ as $x \to \infty$, while the difference x itself diverges.

Example 5.1 (Stirling's formula)**.**

$$n! \sim \sqrt{2\pi n}\left(\frac{n}{e}\right)^n \qquad (n \to \infty).$$

The ratio of the two sides tends to 1, even though their difference grows without bound — the hallmark of $\sim$ as distinct from a literal approximation.

6 Comparing growth at infinity

A standard hierarchy of orders of infinity, each term o of the next as $x \to \infty$ (for constants $p > 0$, $b > 1$, and $0 < c < 1$):

$$1 \prec \ln x \prec x^{\varepsilon} \prec x^p \prec x^p \ln x \prec x^{p+\varepsilon} \prec e^{x^c} \prec b^x \prec x! \prec x^x,$$

where $f \prec g$ abbreviates $f = o(g)$, $\varepsilon > 0$ is arbitrarily small, and $b_1^x \prec b_2^x$ whenever $1 < b_1 < b_2$. This hierarchy is the usual first resort for limit-comparison arguments (e.g. deciding convergence of $\int^{\infty} f$ or $\sum f(n)$ by comparing f against a power of x or an exponential).

Example 6.1. $\lim_{x\to\infty} \dfrac{\ln x}{x^{0.01}} = 0$ follows immediately from $\ln x \prec x^{\varepsilon}$ above, without L'Hôpital.

7 Common pitfalls

- **Wrong direction of comparison.** $f = O(g)$ bounds f *above* in size; it says nothing about $g = O(f)$. Confusing "at most the order of" with "the same order as" is the single most common error.
- **O does not imply the limit of the ratio exists.** $\sin x = O(1)$ as $x \to \infty$, but $\sin x$ has no limit and $\sin x \not\to c$ for any constant c.
- **The base point matters.** $x^2 = O(x)$ fails as $x \to \infty$ but holds as $x \to 0$; always state (or infer from context) which limit is in force before writing a Landau expression.
- **$o(1)$ and $O(1)$ are shorthand for "tends to 0" and "stays bounded."** These are the two most frequent instances and worth recognizing on sight: $f = o(1) \iff f(x) \to 0$; $f = O(1) \iff f$ is bounded near a.
- **Landau notation is not a substitute for a rigorous bound when one is needed.** It is a bookkeeping device for tracking leading-order behavior; when a problem asks for an explicit constant or an explicit neighborhood, the O has to be unpacked back into its ε–δ (or C–neighborhood) form.